Kishan K

AI/ML Software Engineer | GenAI, Computer Vision & Backend Systems

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PROFESSIONAL EXPERIENCE

Software Engineer (AI/ML) 10/2025 – Present

Silo Fortune Pvt Ltd Bengaluru - On Site

- Fine-tuned **Qwen-2.5-1.5B** (4-bit QLoRA, r=16) on **178,980** knowledge-base-generated advisory pairs covering **312 crops**, **1,693 diseases** and **12 Indic languages**; paired it with **Gemini 2.5** diagnostics to return 5-stage disease progression and revenue-loss estimates per crop.
- Engineered a price-forecasting service for **6,273 mandis** and **346 commodities**: a 6-model ensemble (**CatBoost, LightGBM, XGBoost, GBR, RF, SVR**) with lag and sin/cos seasonality features producing 12-week projections; rolling-origin backtest on 15 mandi–commodity series (59 origins): **13.4%** MAPE vs. 17.0% for a naive baseline.
- Designed a two-level report cache (report template keyed on 5 bucketed satellite metrics, plus per-language translation store in **PostgreSQL**) behind the **FastAPI** agro-monitoring service: **99.4%** template hit rate and ∼**97%** fewer **Gemini** calls (8,130 hits vs. 52 generations, production counters).
- Contributed to satellite processing (Sentinel-2/Landsat NDVI/EVI/NDWI) and an **ISRIC** soil-based crop recommendation engine; services deployed with **Docker**, **Jenkins** and **AWS ECS Fargate**. The platform reached **50,000+** installs and **3,500+** subscribers (admin-panel metrics).

Applied ML Engineer (Part-Time Contract) 09/2026 – Present

Alpify Technologies LLP Remote

- Developed a lead-scoring pipeline: ETL unifying 611 leads from 12 sheets across 6 channels (Meta, Google, outbound, website), plus an **XGBoost** contactability model (5-fold CV **AUC 0.81**, 95% CI 0.76–0.83, n=611) that assigns P1/P2/P3 call-priority tiers.

TECHNICAL SKILLS

GenAI & LLM Engineering: Google Gemini 2.5, Qwen-2.5, Fine-Tuning (QLoRA, Unsloth, PEFT), RAG, LangGraph, LangChain, pgvector, RAGAS, Prompt Engineering, Structured JSON Outputs

AI/ML & Computer Vision: YOLOv8, DeepSORT, OpenCV, PyTorch, Time-Series Forecasting, Ensembles (CatBoost, LightGBM, XGBoost, Scikit-Learn), Remote Sensing (Sentinel-2, Rasterio)

Software & Backend Engineering: FastAPI, Django, Flask, GraphQL (Strawberry), AsyncIO, REST APIs, Microservices, Celery, RabbitMQ, Pydantic, SQLAlchemy, Alembic

Cloud, DevOps & Databases: Docker, AWS (ECS Fargate, S3, RDS), Jenkins CI/CD, Git, Linux, PostgreSQL, Redis, Neo4j (Cypher), Python, SQL

KEY PROJECTS

for more projects: kishan-k.pages.dev/#projects

Sentinel – Agentic Financial Research & Regulatory Compliance Engine GitHub | Hugging Face

- Adapted **Qwen-2.5-7B Instruct** (4-bit **QLoRA**, r=16) into a 4-class SEBI-compliance classifier on 2,174 Gemini-generated claims: **270/271** correct (**99.6%**, 95% CI 98–99.9%) on a 271-claim held-out split with no train/test duplicates.
- Architected a **LangGraph** research → compliance agent over hybrid retrieval (BM25 + **pgvector**, reciprocal-rank fusion, BGE cross-encoder rerank) with automated SEBI rewriting; **RAGAS** on 100 queries: **0.88** faithfulness, **0.85** answer relevancy.
- *Tech Skills: LangGraph, Qwen-2.5, QLoRA (Unsloth, PEFT), RAGAS, pgvector, FastAPI, PostgreSQL, Docker*

Kisaan Sampurna – Multimodal Agri-Advisory & Market Intelligence Google Play Store | GitHub

- Prototyped a **Neo4j** + **LangGraph** + **Strawberry GraphQL** agri-advisory copilot (entity extraction → graph traversal → advisory synthesis) over a 3,140 crop–disease knowledge base, with a **RabbitMQ** task queue and an in-memory fallback graph; using company dev data as a proof of concept.
- Served multilingual advisories through **FastAPI** on **AWS ECS Fargate**; on 300 validation prompts the domain-specialised 1.5B model produced the required 4-section advisory format **300/300** (base model: 0/300), matching the action field **99.3%** and infection-rate range **85.7%** of the time (95% CI 81–89%).
- *Tech Skills: Google Gemini 2.5, Qwen-2.5, QLoRA (Unsloth), Neo4j, Cypher, LangGraph, GraphQL, CatBoost, LightGBM, FastAPI, PostgreSQL, Redis, RabbitMQ, AWS (ECS, S3)*

Gau Swastha – Cattle Semen Motility Analysis & Livestock Advisory Backend *Client Demo*

- Implemented an asynchronous semen-analysis microservice: **YOLOv8** sperm detection with **DeepSORT** tracking on microscopy video, classifying tracks as progressive / non-progressive / immotile with a motility quality score; **FastAPI** upload API (202 + report id) feeding a **Celery** worker on a strict-FIFO **Redis** queue (single-task, memory-bounded for a 2 vCPU / 4 GB container), with **S3** video storage and **PostgreSQL** reports; demoed to clients.
- Delivered the structured-API backend (Gau-Gopalan-Seva) with cached GenAI responses and multilingual knowledge-base translations for the mobile app, and extended the WhatsApp/third-party AI orchestration layer with multilingual weather advisories.
- *Tech Skills: YOLOv8, DeepSORT, OpenCV, PyTorch, Google Gemini, FastAPI, Celery, Redis, AWS (S3, ECS), PostgreSQL, Docker*

OPEN SOURCE CONTRIBUTIONS

NotifKit – Open-Source Notification Infrastructure notifkit.dev | GitHub

- Contributor (3 commits) to a self-hosted notification platform whose single `notify()` call handles push, email, SMS, and webhooks with fallback, retries, and durable workflows. *(Node.js, TypeScript, PostgreSQL, Redis Streams)*

EDUCATION

B.E. in Artificial Intelligence and Machine Learning 12/2021 – 07/2025

Ramaiah Institute of Technology, Bengaluru **CGPA: 8.6 / 10**

PUBLICATIONS

Performance Evaluation of Deep Learning Models for Predicting Alzheimer's Disease *IEEE, 2024*

Enhancing Predictive Maintenance with SHAP and LIME *ICAI – ARSSS, 2025*